

2018 Year 6 Chemistry Practical Examination 2

List of Chemical

Question 1			
<i>Description</i>	<i>Label (if any)</i>	<i>Quantity per student</i>	<i>Remarks</i>
0.030 mol dm ⁻³ acidified iron(III) chloride Dissolve 8.11 g of FeCl ₃ .6H ₂ O [C] in each dm ³ of 0.20 mol dm ⁻³ hydrochloric acid. 0.20 mol dm ⁻³ hydrochloric acid made by diluting 2.0 mol dm ⁻³ hydrochloric acid [H] (preparation instructions on page 63 of the 2015 syllabus) 10-fold.	FA 1	150 cm ³	1 bottle per student; to be placed on each student bench.
0.060 mol dm ⁻³ potassium iodide Dissolve 9.96 g of KI in each dm ³ of solution.	FA 2	80 cm ³	
0.0060 mol dm ⁻³ sodium thiosulfate Dissolve 29.78 g of Na ₂ S ₂ O ₃ .5H ₂ O in each dm ³ of solution. Dilute this solution 20-fold.	FA 3	150 cm ³	
20 g dm ⁻³ starch solution	Starch indicator	80 cm ³	

The time t should be in the range 15 to 20s for experiment 1.

Question 2			
<i>Description</i>	<i>Label (if any)</i>	<i>Quantity per student</i>	<i>Remarks</i>
Powdered anhydrous CaCO ₃ (fresh chemicals should be used, chemicals after packed in small bottle should be kept air-tight.)	FA 4	About 1 gram, cap in a small weighing bottle.	1 bottle per student; to be placed on each student bench.
2.00 mol dm ⁻³ hydrochloric acid, HCl	FA 5	70 cm ³	

0.130 mol dm ⁻³ sodium hydroxide, NaOH (labelled as 0.140M in the qn paper)	FA 6	150 cm ³	
Thymolphthalein indicator *Freshly prepared 0.04 g thymolphthalein in 50 mL of 95% ethanol. Dilute this solution to 100 mL with distilled water.	thymolphthalein	5 cm ³	

Question 3			
<i>Description</i>	<i>Label (if any)</i>	<i>Quantity per student</i>	<i>Remarks</i>
manganese(IV) oxide, MnO ₂ , in a capped small bottle	FA 8	6 g	1 set per student; to be placed on each student bench.
approximately 0.5 mol dm ⁻³ sulfuric acid, H ₂ SO ₄	FA 9	40 cm ³	
sodium ethanedioate, Na ₂ C ₂ O ₄ , in a capped small bottle	FA 10	2.8 g	
approximately 0.2 mol dm ⁻³ manganese(II) sulfate, MnSO ₄ , made by dissolving, in deionised water, about 84.5 g of MnSO ₄ .H ₂ O in each dm ³ of solution. This solution must be stored in a capped bottle	FA 11	20 cm ³	

Others

Bench reagents			1 bottle per student; to be placed on each student bench. *Top up NaOH(aq), NH ₃ (aq), limewater after each shift
Deionised water	-	1 large bottle	Each student requires about 500 cm ³ , so they will top up at least once during the examination.

List of Apparatus

<i>Description</i>	<i>Quantity</i>	<i>Remarks</i>
100 cm ³ conical flask, labelled X	1	
Burette	1	Clamped with retort stand. To check the tips and replaced if chipped.
Pipette 25.0 cm ³	1	
Filter funnel	2	
250 cm ³ volumetric flask	1	
250 cm ³ conical flask	2	
Pipette filler	1	
White tile	1	
25 cm ³ measuring cylinder	2	
10 cm ³ measuring cylinder	2	
Alcohol thermometer graduated to 0.5°C	1	
Ignitor	1	
Tripod stand + wire gauze	1	
Test-tube rack + test tube holder + 1 large and 1 small brushes	1	
Boiling tube	1	
large test-tube	5	
Small test-tube	3	
Delivery tube	1	
stopwatch	1	
100 cm ³ beaker	2	
plastic teat pipettes	8	
Small spatula	1	
Glass rod	1	
Paper towel	6	
Round filter paper	2	To be placed in small ziplock bag
red, blue litmus paper strips	4 each	
Electronic weighing balance	See remarks	5/6 per lab; 1 waste beaker beside each balance
Small weighing bottle with cap	1	
Permanent marker	1	
Bunsen burner	1	

Note: Place the following on the Teacher's Bench

- 6 additional sets of apparatus
- 10 additional sets of chemicals